

An Explicit Closed-Form G_2 Ansatz on a $K3$ -Coassociative Neck with Hyperkähler Rotation and Picard–Lefschetz Wirtinger Certificate

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Abstract

We describe an explicit closed-form ansatz for a positive 3-form φ on a $K3$ -coassociative neck model in cohomogeneity-1, organized around a determinant-preserving radial scale family $(a(t), b(t))$ with $\det(g) = 65/32$ exact, a smoothly rotating hyperkähler triple $\Omega_i(t) = R_{ij}(t)\Omega_j^0$ with $R(t) \in \mathrm{SO}(3)$, and a variable base coframe with structure constants $c_{i,jk}(t)$ whose chosen values cancel the rotation-induced $d\varphi$ components in the reduced $\theta_{ij} \wedge \Omega_k$ block. A reduced Chebyshev solver produces a real, positive-definite metric for which the determinant constraint, the $d\varphi$ block, and the $d \star \varphi$ scale block close to machine precision ($\leq 10^{-12}$ in all primary blocks).

A calibrated Fano-meridian branch (with $\int_0^1 \nu(t) dt = \pi$ along a chosen Fano axis) integrates the rotation $R(t)$ along the neck to an order-two element of $\mathrm{SO}(3)$ matching the rank-one Picard–Lefschetz reflection of a $K3$ -coassociative monodromy to error $\approx 1.2 \times 10^{-14}$. The implicit discriminant graph of the fibration is identified with the seven-component Fano Hopf link in S^3 (seven distinct fibers of the Hopf fibration with pairwise linking number $+1$), and a five-layer symbolic certificate verifies that, after Wirtinger reduction, the resulting integer relation matrix matches the Donaldson cellular complex used in the Joyce–Karigiannis $\mathbb{Z}_2^3$ construction [5], with torsion-free cokernel $\mathbb{Z}^3$.

The positive G_2 -structure ansatz is consistent with, but does not prove, the existence of a smooth compact G_2 structure with Betti numbers $(b_2, b_3) = (21, 77)$. Conditional on the Joyce–Karigiannis $\mathbb{Z}_2^3$ orbifold resolution and analytic gluing theorem [5], the ansatz provides explicit closed-form data on the cohomogeneity-1 trace of the resulting compact manifold, complementary to the Newton–Kantorovich certificate of [A] and the spectral analysis of [B]. The remaining open analytical questions are an explicit smooth coframe on $S^3 \setminus \Gamma_{\mathrm{Fano}}$ and interval-arithmetic certification of the residuals. These are not blockers for the reduced closed-form ansatz and symbolic certificate reported here; they are, however, necessary upgrades for a fully global compact torsion-free G_2 metric statement.

Keywords: G_2 holonomy, $K3$ -coassociative fibration, hyperkähler rotation, base coframe, Picard–Lefschetz monodromy, Hopf fibration, Wirtinger presentation, Newton–Kantorovich.

1 Introduction

1.1 The GIFT topological signature

Throughout this note we work with the *GIFT 7-manifold target*: a compact, simply connected 7-manifold K_7 with G_2 holonomy and Betti numbers

$$(b_2(K_7), b_3(K_7)) = (21, 77).$$

This pair does not appear among the catalogued TCS examples [6, 1] but is the natural GIFT signature derived from the algebraic identity

$$b_2 = N_{\text{gen}} \cdot \dim(K_7), \quad b_3 = (\text{rank } E_8 + N_{\text{gen}}) \cdot \dim(K_7),$$

with $\dim(K_7) = 7$, $\text{rank } E_8 = 8$, $N_{\text{gen}} = 3$.

1.2 What this note provides

The body of the note describes:

- A determinant-preserving radial scale family $(a(t), b(t))$ with $\det(g) = 65/32$ exact (§3), parameterised by a single Chebyshev profile $u(t)$.
- A smoothly rotating hyperkähler triple $\Omega_i(t) = R_{ij}(t) \Omega_j^0$ with $R(t) \in \text{SO}(3)$ (§4), integrated by a Lie-group midpoint Euler scheme with SVD reprojection.
- A variable base coframe with structure constants $c_{i,jk}(t)$ (§4) whose chosen values cancel the rotation-induced $d\varphi$ components in the reduced $\theta_{ij} \wedge \Omega_k$ block.
- A calibrated Fano-meridian branch (§5) reproducing the rank-one Picard–Lefschetz reflection of a $K3$ -coassociative monodromy to error $\approx 1.2 \times 10^{-14}$.
- An identification of the implicit discriminant graph as the seven-component Fano Hopf link in S^3 (§5), with deterministic projection yielding 54 transverse double-point crossings.
- A five-layer symbolic certificate (§6) that the resulting Wirtinger presentation has integer cokernel $\mathbb{Z}^3$, torsion-free, matching the Donaldson cellular complex of the Joyce–Karigiannis $\mathbb{Z}_2^3$ construction [5].

The full closed-form positive G_2 -structure ansatz is

$$\varphi(t) = a(t)^3 \theta_{123} + a(t) b(t)^2 \sum_{i=1}^3 \theta_i \wedge \Omega_i(t).$$

1.3 What this note does *not* claim

The note does *not* prove:

- A globally smooth metric on a compact 7-manifold. The closed-form ansatz operates at the cohomogeneity-1 neck level; the smooth compact statement requires the Joyce–Karigiannis 2017 analytic gluing theorem [5] applied to the Joyce–Karigiannis $\mathbb{Z}_2^3$ topological data [5].
- Interval-arithmetic bounds on the analytic residuals. Residuals are reproducible to floating-point precision but not yet wrapped in rigorous interval bounds.
- An explicit smooth global coframe for $S^3 \setminus \Gamma_{\text{Fano}}$. We provide the discrete topological data (linking matrix, crossing signs, holonomy assignment); the smooth coframe one-forms remain open.

These three open questions are not blockers for the reduced closed-form ansatz and symbolic certificate reported here; they are necessary upgrades for a fully global compact torsion-free G_2 metric statement.

1.4 Document layout

§2 recapitulates the topological setting (Joyce–Karigiannis $\mathbb{Z}_2^3$ orbifold, briefly; see [5] for the orbifold construction). §3 specifies the determinant-preserving radial family. §4 introduces the hyperkähler rotation and base-coframe absorption. §5 describes the Picard–Lefschetz calibration and the seven-Hopf-link spatial embedding. §6 verifies the symbolic Wirtinger certificate in five layers. §7 records honest scope, comparisons with [A] and the JK construction, and open questions.

Each section produces deterministic certificates reproduced by the Python audit `gift_core.examples.verify_donaldson` (73 PASS checks). The relevant Lean modules are `GIFT/Foundations/JoyceKarigiannisConstruction.lean` and `GIFT/Foundations/DonaldsonGlobalBaseAudit.lean`. No new axioms beyond the existing 15 (4 main-chain + 11 interval-arithmetic) inherited from [A] are introduced.

2 Topological Existence — Joyce–Karigiannis $\mathbb{Z}_2^3$

2.1 Setup

The Joyce–Karigiannis (JK) 2017 framework [5] resolves orbifolds $T^3 \times K3/G$ where $G \subset \text{Aut}(T^3) \times \text{Aut}(K3)$ is a finite group acting with A_1 -type singularities. Each fixed component is resolved by Eguchi–Hanson gluing, yielding a smooth compact 7-manifold with a torsion-free G_2 structure (under the analytic gluing hypothesis of [5]).

For $G = \mathbb{Z}_2^3 = \langle s_1, s_2, \tau \rangle$ with the mixed-parity action

- $s_1, s_2 \in V_4$: symplectic involutions on $K3$, sign flips on T^3 ;
- τ : non-symplectic (anti-symplectic) involution on $K3$, free on T^3 ;

the resulting fixed-locus components are 12 copies of T^3 (from V_4 orbits), 1 copy of $S^1 \times \Sigma_2$ (from τ , where Σ_2 is a genus-2 curve), 2 copies of $S^1 \times \mathbb{CP}^1$ (rational curves of τ), 3 copies of $S^1 \times T^2$ (from $s_i\tau$, with fixed-locus shape $(g, k) = (1, 1)$), and 3 more copies of $S^1 \times \mathbb{CP}^1$ (rational curves of $s_i\tau$). Total: 21 components, with $b_0 = 21$ and $b_1 = 55$.

The JK Betti formula gives:

$$\begin{aligned} b_2(N) &= b_2(T^3 \times K3/\mathbb{Z}_2^3) + b_0(\text{fixed}) = 0 + 21 = 21, \\ b_3(N) &= b_3(T^3 \times K3/\mathbb{Z}_2^3) + b_1(\text{fixed}) = 22 + 55 = 77, \\ \chi(N) &= 0 \quad (\text{Poincaré duality, 7-dim}). \end{aligned}$$

2.2 Lattice-theoretic existence

The required $\mathbb{Z}_2^3$ action on a $K3$ admitting the polarisation degree $\eta^2 = 8$ exists abstractly:

- [7]: any $V_4 \subset M_{23}$ acts on a $K3$ by symplectic automorphisms. Take $V_4 = \langle s_1, s_2 \rangle$.
- [4]: a non-symplectic involution τ on a $K3$ with discriminant data (r, a, δ) exists iff $a > 16 - r$. Both $(11, 7, 1)$ (giving $(g, k) = (2, 2)$ fixed-locus shape) and $(11, 9, 1)$ (giving $(g, k) = (1, 1)$) satisfy this; combined, they realise the four certified curve cases needed for the active JK ledger.

The Nikulin involution $\sigma_1 = E_8$ -swap on $\Lambda = U^3 \oplus E_8(-1)^2$ is order-2, trace 6, with eigenspace dimensions $(14, 8)$, and coinvariant lattice $T_{\sigma_1} \cong E_8(-2)$ of discriminant 2^8 [8].

2.3 Lean witness

The four phases are encoded in `JoyceKarigiannisConstruction.lean`: phase 1 V_4 symplectic screen ($24 \text{ K3-fixed} \rightarrow 12 \text{ V4-orbits} \rightarrow 12 \text{ T}^3$); phase 2 anti-symplectic obstruction ($\det(\tau)/\det(R) \equiv 1$); phase 2b lattice screen (Nikulin σ_1 + Mukai V_4 + Garbagnati–Sarti); phase 4 Betti formula. The master theorem `jk_z23_construction_realizes_gift_betti` proves $\text{phase4.b2N} = b_2$ and $\text{phase4.b3N} = b_3$ by `native_decide`.

2.4 Honest scope at this level

The JK route certifies *topological existence* of N with the GIFT Betti pair, plus *abstract existence* of the required $K3$ surface with the $\mathbb{Z}_2^3$ action (via Mukai + Garbagnati–Sarti). It does not provide an explicit polynomial coordinate model for the action on a specific $K3$ in moduli (this is deferred to specialists) or a smooth metric (deferred to [5] analytic gluing).

3 The Determinant-Preserving Radial Family

3.1 Ansatz

In an orthonormal local frame $(e_1, \dots, e_7)$ where (e_1, e_2, e_3) spans the base direction and (e_4, e_5, e_6, e_7) spans the $K3$ fibre direction, write the standard G_2 3-form

$$\varphi = a^3 \theta_{123} + a b^2 \sum_{i=1}^3 \theta_i \wedge \Omega_i,$$

with θ_i the dual base coframe and Ω_i a hyperkähler triple on the $K3$ fibre. The induced metric is diagonal:

$$g = \text{diag}(a^2, a^2, a^2, b^2, b^2, b^2, b^2), \quad \det(g) = a^6 b^8.$$

3.2 GIFT determinant constraint

The GIFT framework imposes $\det(g) = 65/32$ exactly. Let $\alpha := (65/32)^{1/14}$. The exponential family

$$a(t) = \alpha e^{4u(t)}, \quad b(t) = \alpha e^{-3u(t)},$$

satisfies $a^6 b^8 = \alpha^{14} e^{(24-24)u} = 65/32$ exactly, for any choice of profile $u(t)$. The exponents $(4, -3)$ are the unique pair (up to sign) such that $6 \cdot 4 + 8 \cdot (-3) = 0$.

3.3 Chebyshev radial profile

The free function $u(t)$ on the neck domain $t \in [-1, 1]$ is parametrised as a Chebyshev expansion with a quartic boundary envelope:

$$u(t) = (1 - t^2)^p \sum_{k=0}^n c_k T_k(t), \quad p = 2.$$

The envelope ensures $u(\pm 1) = u'(\pm 1) = 0$, i.e. cusp matching to flat ends. The default profile minimises $\|u'\|_2$ subject to a fixed centre amplitude $u(0) = u_0$ via a deterministic quadratic program (KKT system, $O(n)$ unknowns).

3.4 Numerical determinant error

At the default $u_0 = 0.0525$, $n = 8$:

$$\max_{t \in [-1, 1]} |\det(g(t)) - 65/32| \approx 3.6 \times 10^{-15},$$

i.e. machine precision. This is *exact* by construction up to floating-point rounding.

4 Hyperkähler Rotation and Base-Coframe Absorption

4.1 The signed-curvature obstruction

A symmetric $SO(3)$ connection $A_i = q_i(t) \theta_i$ with $q_1 = q_2 = q_3 = q(t)$ contributes a curvature $F_i = q^2(t) \theta_{jk}$ which cancels the $d\varphi$ block residual when $q^2 = k(t) := 2u'(t)$ (up to a constant factor). This requires $k(t) \geq 0$ everywhere; for natural neck profiles, $u'(t)$ changes sign on roughly half the domain (47.7% of $[-1, 1]$ at $u_0 = 0.0525$), leaving a signed obstruction.

4.2 Rotated triple

Promote the fixed hyperkähler triple

$$\Omega_1^0 = e^{45} + e^{67}, \quad \Omega_2^0 = e^{46} - e^{57}, \quad \Omega_3^0 = e^{47} + e^{56}$$

to a smoothly rotating triple $\Omega_i(t) = R_{ij}(t) \Omega_j^0$ with $R(t) \in \text{SO}(3)$. The rotation is parametrised by an angular-velocity profile $\nu(t) \in \mathbb{R}^3$ via $\omega(t) = -\epsilon_{ijk} \nu_k(t)$ and integrated as a Lie-group midpoint Euler sequence with SVD reprojection at each step. With step $h = 1/256$, the computed $R(t)$ satisfies $|\det R(t) - 1| < 10^{-12}$ and $\|R^T R - I\|_\infty < 10^{-12}$ on the full domain.

4.3 Rotation-induced $d\varphi$ residual

The rotation introduces new terms in $d\varphi$:

$$d\varphi^{\text{rot}} = ab^2 \sum_i \theta_i \wedge d\Omega_i(t), \quad d\Omega_i(t) = \epsilon_{ijk} \nu_k(t) \Omega_j(t) dt.$$

In the cohomogeneity-1 reduction with $dt = \theta_1$, the four non-trivial 4-form components are

$$\begin{aligned} \text{coeff}(\theta_{12} \wedge \Omega_1) &= +ab^2 \nu_3, & \text{coeff}(\theta_{12} \wedge \Omega_3) &= -ab^2 \nu_1, \\ \text{coeff}(\theta_{13} \wedge \Omega_1) &= -ab^2 \nu_2, & \text{coeff}(\theta_{13} \wedge \Omega_2) &= +ab^2 \nu_1. \end{aligned}$$

4.4 Base coframe absorption

Promote the base (θ_2, θ_3) from flat to non-trivial via structure constants:

$$d\theta_i = c_{i,jk}(t) \theta_j \wedge \theta_k.$$

Choose:

$$\begin{aligned} c_{1,12} &= -\nu_3, & c_{1,13} &= +\nu_2, & c_{1,23} &= 0, \\ c_{2,12} &= 0, & c_{2,13} &= -\nu_1, & c_{2,23} &= 0, \\ c_{3,12} &= +\nu_1, & c_{3,13} &= 0, & c_{3,23} &= 0. \end{aligned}$$

The chosen structure constants cancel the rotation-induced $d\varphi$ components in the reduced $\theta_{ij} \wedge \Omega_k$ block. A full geometric realization requires a smooth coframe satisfying the corresponding integrability conditions. The local Bianchi residual $d^2\theta_i$ is quadratic in ν and lives in θ_{123} , *orthogonal* to the $\theta_{ij} \wedge \Omega_k$ basis where $d\varphi$ residuals live; it is a higher-order correction not affecting the primary closure of the reduced model.

4.5 Joint solver result

For reference parameters $u_0 = 0.0525$, $\nu_{\text{amplitude}} = 0.035$, axis $(1, 0, 0)$, Chebyshev degree 8 for u and 4 for each ν_i :

The metric remains real and positive-definite throughout. No complexification, no indefinite signature.

Quantity	Value
$\det(g) - 65/32$	$< 10^{-14}$
$\ R^T R - I\ _\infty$	$< 10^{-12}$
$ \det R - 1 $	$< 10^{-12}$
$\nu(\pm 1)$	$< 10^{-12}$
Combined $d\varphi$ residual	$< 10^{-12}$
$d \star \varphi$ scale-block residual (ansatz blocks only)	$< 10^{-12}$
Sign coverage (negative u' region)	100% (vs 47.7% rejected by symmetric branch)

Table 1: Joint solver residuals for the rotating-coframe ansatz.

5 Picard–Lefschetz Holonomy and the Seven-Hopf-Link Embedding

5.1 Calibrated meridian rotation

The angular-velocity profile $\nu(t)$ admits a *calibrated meridian branch* in which the integrated rotation along the radial direction returns to a specific element of $\mathrm{SO}(3)$ matching the rank-1 Picard–Lefschetz reflection of the $K3$ -coassociative monodromy.

For axis $(1, 0, 0)$, calibrate the profile so that

$$\int_0^1 \nu_1(t) dt = \pi.$$

Both endpoints $R(\pm 1)$ then land on the same order-two element of $\mathrm{SO}(3)$ (rotation by $\pm\pi$ around the chosen axis = the axis reflection). Numerical results:

Quantity	Value
Endpoint angle	π (exact by calibration)
$R(-1)$ vs target holonomy	error $\approx 1.2 \times 10^{-14}$
$R(+1)$ vs target holonomy	error $\approx 1.2 \times 10^{-14}$
Order-two test $R^2 = I$	error $\approx 2.5 \times 10^{-16}$
Combined rotation + base $d\varphi$	0 (exact in the reduced model)

Table 2: Calibrated meridian branch matches the target Picard–Lefschetz holonomy.

5.2 The spatial embedding question

For the rotation calibration of §5.1 to extend to a *global* construction, the underlying discriminant graph Γ_{Fano} must be realised as an explicit spatial graph in S^3 , with crossing data from a chosen projection so that a Wirtinger presentation of $\pi_1(S^3 \setminus \Gamma_{\mathrm{Fano}})$ [3] can be written down.

A preliminary audit established that the abstract Fano incidence triples (p_a, p_b, p_c) — without spatial embedding data — are *insufficient*. Concretely, naïve line relators $\prod_{p \in L} m_p = 1$ for each Fano line L do not all close as identities under the rank-1 PL representation; line generators $\ell_L = \prod_{p \in L} m_p$ give order-2 elements for 6 of the 7 Fano lines, but the line $(3, 4, 5)$ produces a rotation by $\pi/2$ (order 4), not an order-2 reflection. The geometric reason: products of three reflections in $\mathrm{O}(3)$ are order-2 only when the three

reflection axes are coplanar; for the $(3, 4, 5)$ line in the abstract incidence picture, the axes are not coplanar.

5.3 Seven-component Fano Hopf link

A spatial-embedding audit tested three candidate spatial embeddings:

Candidate	Vertices	Edges	Abelianisation	Verdict
K_7 Fano-coloured	7	21	$\mathbb{Z}^{15}$	obstructed
Heawood incidence graph	14	21	$\mathbb{Z}^8$	obstructed
7-component Fano Hopf link (7 components, S^1)	—	—	$\mathbb{Z}^7$	partial candidate

Table 3: Spatial embedding candidates and their abelianisation ranks.

The winner: *seven distinct fibres of the Hopf fibration* $S^3 \rightarrow S^2$, with all pairwise linking numbers $+1$. The complement has fundamental group

$$\pi_1\left(S^3 \setminus \bigcup_{i=1}^7 F_{p_i}\right) = F_6 \times \mathbb{Z}$$

(trivial S^1 -bundle over the punctured sphere $S^2 \setminus \{p_1, \dots, p_7\}$). The abelianisation is $\mathbb{Z}^7$, matching the 14-oriented-meridian presentation after orientation pair identifications.

The seven Fano lines of the abstract incidence structure correspond to the seven *components*, not to triples of points; the line $(3, 4, 5)$ obstruction is resolved because each line carries its own meridian (a single reflection on the moved $K3$ direction) rather than being a derived product of three point-meridians.

5.4 Quaternionic embedding

Concretely, label the seven Hopf fibres by seven unit vectors $v_i \in \mathbb{C}^2 \cong \mathbb{H}$:

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (0, 1), \\ v_3 &= (1, 1)/\sqrt{2}, & v_4 &= (1, i)/\sqrt{2}, \\ v_5 &= (1, -1)/\sqrt{2}, & v_6 &= (1, -i)/\sqrt{2}, \\ v_7 &= (1, e^{i\pi/4})/\sqrt{2}. \end{aligned}$$

Each v_i defines a Hopf fibre

$$F_{v_i} = \{e^{i\theta} v_i : \theta \in [0, 2\pi)\} \subset S^3 \subset \mathbb{C}^2.$$

Pairwise linking numbers are $+1$ (computed from the projective distance $\sqrt{1 - |\langle v_i, v_j \rangle|^2}$ which is > 0 for $v_i \nparallel v_j$).

5.5 Deterministic projection

A deterministic stereographic projection (with pole and frame chosen to maximise transverse double-point separation) yields:

This is the explicit input to the symbolic Wirtinger reduction in §6.

Quantity	Value
Crossing count	54
Min XY separation	$\approx 2.9 \times 10^{-3}$
Min Z gap (over/under)	≈ 0.276
Has transverse double points only	true
Smooth oriented projection certified	true

Table 4: Deterministic projection of the seven-Hopf-link.

6 Symbolic Wirtinger / Tietze Certificate

6.1 Five-layer structure

The certificate establishes that the seven-Hopf-link Wirtinger presentation reduces, after Tietze moves, to exactly the 14-meridian / 11-relator presentation of the Donaldson cellular complex (`FanoMeridianModel`), with torsion-free integer cokernel $\mathbb{Z}^3$.

The certificate is structured in five layers, all deterministically verified:

Layer 1 — Topology

For seven distinct fibres of the Hopf fibration $S^3 \rightarrow S^2$, the complement is

$$S^3 \setminus \bigcup F_{p_i} \cong (S^2 \setminus \{p_1, \dots, p_7\}) \times S^1,$$

a trivial S^1 -bundle over the punctured sphere (the Euler class restricted to the punctured surface vanishes). Hence

$$\pi_1(S^3 \setminus \bigcup F_{p_i}) = F_6 \times \mathbb{Z},$$

with abelianisation $\mathbb{Z}^7$.

Layer 2 — Algebraic relation matrix

The 14×11 integer relation matrix of the Donaldson cellular complex consists of:

- 7 orientation rows: $m_i^+ - m_i^- = 0$ for $i = 1, \dots, 7$.
- 4 projective rows: integer kernel basis of the map $\mathbb{Z}^7 \rightarrow \mathbb{Z}^3$ given by the seven Fano points in $(\mathbb{F}_2)^3$.

Computational properties:

- rank = 11, cokernel free rank = 3;
- gcd of maximal minors = 1 (torsion-free cokernel);
- 232 nonzero maximal minors.

Layer 3 — Smith normal form

The torsion-free cokernel (Layer 2) plus the standard fact that the gcd of $r \times r$ maximal minors equals $\prod_{i=1}^r d_i$ (where d_i are the Smith invariant factors) implies all eleven invariant factors equal 1. Hence the cokernel decomposes as

$$\text{coker}(\text{relation matrix}) = \mathbb{Z}^3 \oplus 0,$$

with no torsion. This is exact, not numerical.

Layer 4 — Compatibility with abelian-target representations

The map $\pi_1 \rightarrow \pi_1^{\text{ab}} = \mathbb{Z}^7$ followed by the quotient $\mathbb{Z}^7 \rightarrow \mathbb{Z}^3$ factors any representation $\rho : \pi_1 \rightarrow G$ with abelian image through the Donaldson cellular group, by universal property.

Layer 5 — Picard–Lefschetz lattice witness

A concrete family of seven (-2) -classes in $T = U^2 \oplus E_8(-1)^2 \oplus \langle -8 \rangle$ realizing the four Fano projective relations as additive lattice equations: parametrise

$$\beta(p) = \sum_{i:p_i=1} \beta_i \in T, \quad p \in (\mathbb{F}_2)^3 \setminus \{0\},$$

where $(\beta_0, \beta_1, \beta_2)$ are three independent lattice elements with Gram matrix shape $\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$. The four Fano projective relations (as integer kernel rows of the 3×7 Fano-point map) reduce to identities

$$\sum_i k_i \cdot \beta(p_i) = 0$$

where the k_i are the kernel coefficients. All four are *automatically* satisfied by the $\mathbb{F}_2$ -linear parametrisation, verified symbolically (4/4).

6.2 Combined certificate

The five layers together establish: starting from the seven-Hopf-link Wirtinger presentation (Layer 1), the abelianisation reduces to $\mathbb{Z}^7$ (Layer 1); the Donaldson cellular relations cut this to $\mathbb{Z}^3$ with no torsion (Layers 2 + 3); any abelian-target representation factors (Layer 4); and an explicit lattice witness realising the four projective relations exists (Layer 5). The Lean status `globalDonaldsonBaseGeometryStatusCertificate` is upgraded from `compatibleOpen` (preceding release) to `matches`.

Level	Certificate	Lean theorem
Topological existence (JK $\mathbb{Z}_2^3$)	4-phase audit	<code>jk_z23_construction_realizes_gift_betti</code>
Det + $d\varphi + d \star \varphi$ residuals	machine precision	<code>radial_*_compensated</code>
HK rotation + base coframe absorption	machine precision	(Python certs)
Calibrated PL holonomy match	1.2×10^{-14} error	<code>fano_meridian_rotation_matches_*</code>
Spatial embedding identification	7-Hopf link, 54 crossings	<code>at_least_one_spatial_embedding_*</code>
Symbolic Wirtinger certificate (5 layers)	deterministic	<code>fano_seven_link_symbolic_wirtinger_*</code>

Table 5: Summary of certificates and Lean theorems.

7 Honest Scope, Comparison, and Open Questions

7.1 What is established

These layers are reproducible by `python3 -m gift_core.examples.verify_donaldson_direct` (73 PASS), and the underlying theorems compile in Lean (8392 jobs, 15 axioms unchanged, 0 sorry).

7.2 What remains open

Three deferred upgrades, none of which blocks the constructive chain above:

1. **Smooth global $S^3 \setminus \Gamma_{\text{Fano}}$ coframe geometry.** The 54-crossing diagram is explicit; the *smooth metric* on the complement (i.e., the actual coframe one-forms θ_i as smooth global sections) is not yet constructed. Lie-group S^3 candidates (round, Berger, squashed) are obstructed by their antisymmetric Maurer–Cartan structure constants; the actual [2] base must be a non-Lie-group geometry, identifiable from the discriminant data of the $K3$ -coassociative fibration.
2. **Interval-arithmetic certification of analytic residuals.** The $d\varphi, d \star \varphi$ residuals are numerically $< 10^{-12}$; an interval-bound certificate (analogous to the $K3$ `IntervalCertificates.lean` axioms) would upgrade this to a formal mathematical inequality.
3. **Joyce–Karigiannis 2017 analytic gluing for the GIFT $\mathbb{Z}_2^3$ data.** The [5] theorem is stated in general; an explicit application to our chosen $K3 + T^3/\mathbb{Z}_2^3$ data, with verified gluing-radius bounds, would complete the smooth metric existence on N .

7.3 Comparison with Newton–Kantorovich certificate (Paper A)

The Paper A NK certificate [A] establishes existence and local uniqueness of a torsion-free G_2 metric on the same neck model, within an NK ball of radius $\delta g/g \leq 4.86 \times 10^{-6}$ around an explicit Chebyshev–Cholesky parametrisation. The present construction provides:

- A *closed-form* parametric ansatz with all residuals to machine precision (vs Paper A’s numerical 169-parameter optimisation).
- A *global* discriminant identification (the Fano Hopf link in S^3).
- A *symbolic* Wirtinger certificate matching the cellular complex.

The two are complementary: Paper A gives a numerical certificate of existence in a small ball; the present note gives explicit closed-form data for the *centre* of the construction together with a global topological identification.

7.4 Comparison with Joyce–Karigiannis $\mathbb{Z}_2^3$ construction

The Joyce–Karigiannis $\mathbb{Z}_2^3$ construction [5] provides the topological existence proof at the abstract level via orbifold resolution. The present note provides the *explicit parametric ansatz* on the neck of N , together with the spatial embedding of the implicit $K3$ -coassociative discriminant. Together:

- JK route $\rightarrow$ existence of N with $(b_2, b_3) = (21, 77)$ and a smooth torsion-free G_2 structure (modulo [5] gluing).
- Present note $\rightarrow$ explicit closed-form data on the neck of N , with PL holonomy descent through the seven-Hopf-link discriminant.

Whether the explicit ansatz here is the cohomogeneity-1 trace of the JK metric near a fixed-locus component, or is associated with a distinct (Donaldson-type) candidate, is an interesting structural question that we leave open.

Author’s note

This framework was developed through sustained collaboration between the author and several AI systems, primarily Claude (Anthropic), with contributions from GPT (OpenAI), Gemini (Google), and Aristotle (Harmonic) for specific mathematical insights. Architectural decisions and many key derivations emerged from iterative dialogue sessions. This collaboration follows a transparent crediting approach for AI-assisted mathematical research.

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Author’s Related Works

- [A] B. de La Fournière, “A Certified Torsion-Free G_2 Structure on a TCS Neck Model via Computer-Assisted Proof,” Zenodo [10.5281/zenodo.19892350](https://zenodo.org/record/19892350) (2026).
- [B] B. de La Fournière, “Spectral Geometry of an Explicit G_2 Metric: Laplacian Spectrum and Harmonic Forms,” Zenodo [10.5281/zenodo.19893371](https://zenodo.org/record/19893371) (2026).
- [C] B. de La Fournière, “Newton–Kantorovich diagnostics on a Donaldson $K3$ metric: two β estimates, machine-checked inequalities,” Zenodo [10.5281/zenodo.19708916](https://zenodo.org/record/19708916) (2026).

A Reproducibility pipeline

All numerical and symbolic results in this note are reproduced by the GIFT core repository at <https://github.com/gift-framework/core>, release tag v3.4.19:

```
git clone https://github.com/gift-framework/core
cd core
git checkout v3.4.19

# Lean verification (8392 jobs, 0 sorry, 15 axioms)
lake build

# Python audit (73 PASS checks)
PYTHONPATH=contrib/python python3 -m gift_core.examples.verify_donaldson_direct

# Donaldson direct ansatz dense report
PYTHONPATH=contrib/python python3 -m gift_core.examples.donaldson_direct
```

Key Lean modules:

- GIFT/Foundations/JoyceKarigiannisConstruction.lean (§2)
- GIFT/Foundations/DonaldsonCoassociativeFibration.lean (§§3–4)
- GIFT/Foundations/DonaldsonGlobalBaseAudit.lean (§§5–6)

Key Python modules:

- `gift_core/geometry/donaldson.py` (~2700 lines: Donaldson G_2 Ansatz, HyperkahlerRotation, BaseCoframe, FanoSevenComponentLink, spatial embedding audits)
- `gift_core/geometry/wirtinger_symbolic.py` (~290 lines: 5-layer symbolic certificate)

B Reference equations

```
# Determinant-preserving family
alpha  = (65/32)^(1/14)
a(t)   = alpha * exp(4*u(t))
b(t)   = alpha * exp(-3*u(t))
det(g) = a^6 * b^8 = 65/32                                exact

# Hyperkahler rotation
omega(t) = -eps[k] * nu_k(t)                               in so(3)
R'(t)    = omega(t) * R(t),    R(0) = Id                  in SO(3)
Omega_i(t) = R_ij(t) * Omega_j^0                          rotated triple

# Base coframe
dtheta_1 = -nu_3 theta_{12} + nu_2 theta_{13}
dtheta_2 = -nu_1 theta_{13}
dtheta_3 = +nu_1 theta_{12}

# Fano-meridian calibration
int_0^1 nu_axis(t) dt = pi                                 pi-pulse
R(+/-1) = order-2 reflection along Fano axis              error 1.2e-14
R(+/-1)^2 = I                                              error 2.5e-16

# Seven-Hopf-link
v_1 = (1, 0),    v_2 = (0, 1),    v_3 = (1, 1)/sqrt(2),
v_4 = (1, i)/sqrt(2),    v_5 = (1, -1)/sqrt(2),
v_6 = (1, -i)/sqrt(2),    v_7 = (1, exp(i pi/4))/sqrt(2)
F_{v_i} = {exp(i theta) v_i : theta in [0, 2pi]} subset S^3 subset C^2
linking_matrix(F_{v_i}, F_{v_j}) = +1 for all i != j

# Wirtinger / Tietze
14 x 11 relation matrix (FanoMeridianModel):
  7 orientation rows + 4 projective rows
rank = 11
cokernel free rank = 3
cokernel = Z^3 (no torsion)
```

```
Smith invariant factors = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1)

# Picard-Lefschetz lattice witness
beta(p) = sum_{i : p_i=1} beta_i in T,    p in (F_2)^3 \ {0}
4 Fano projective relations satisfied (verified symbolically)
```